

Lesson 4-5 · Period 1 (Quick)

Solving Rational Equations + Chemistry DOK-3 (Q#2/Q#6/Q#7
Prep)

Klimesara-adapted · Student-centered · No Blooket

DOK 1

5 min

bridge from 4-4

Do Now · Nicky vs. Tavon Both students solve $\frac{1}{2}x + \frac{2}{5} = \frac{9}{10}$ — Nicky subtracts fractions; Tavon multiplies both sides by 10.

- (A) What are the advantages/disadvantages of each strategy?
- (B) Are both methods correct?
- (C) Why did Tavon choose to multiply by 10?

DOK 1–2

13 min

teacher models Ex 1 + Ex 3

Rules — Solving Rational Equations

1. Find the LCD of all fractions.
2. Multiply BOTH SIDES by the LCD to clear fractions.
3. Solve the resulting polynomial equation.
4. CHECK each candidate in the ORIGINAL equation — any candidate that makes a denominator zero is **extraneous**.

Domain restrictions are set BEFORE clearing denominators.

Launch — Example 3 (Extraneous Solution)

Algebra 2 · Lesson 4-5 ·

Single-Period Quick

DOK 1–2

13 min

teacher models — COMPRESSED 2 examples

Key idea: why candidates can be extraneous Multiplying both sides by an expression containing x can **introduce** values that are not in the domain of the original equation.

Example 3 pattern:

1. State domain restrictions **first** (values that make any denominator zero).
2. Solve the cleared equation — collect candidate solutions.
3. Test each candidate against the restrictions. Extraneous = throw it out.

Extraneous-solution trap = #1 error on this lesson's assessment items!

Explore — Think-Pair-Share (32 min)

Algebra 2 · Lesson 4-5 · Single-Period

Quick

DOK 2–3

32 min

student work

6 items in order. Plan ~10 min for Practice #33 (Chemistry). Wed F: skip #14 + #25.

Try It 1 · Solve a rational equation What is the solution to each equation?

$$\frac{2}{x+5} = 4 \quad \frac{1}{x-7} = 2$$

Try It 3 · Identify extraneous solution $\frac{1}{x+2} + \frac{1}{x-2} = \frac{4}{(x+2)(x-2)}$ What is the solution?

Practice #10 · Error analysis (Miranda) Describe and correct Miranda's error. What is the correct solution?

Explore (cont.) — Work-Rate + DOK-3 Algebra 2 · Lesson 4-5 · Single-Period Quick

DOK 2–3

32 min

student work

Continue in order. Practice #33 is the big one — plan ~10 min.

Practice #14 · Extraneous trap Solve $\frac{x^2 - 7x - 18}{x + 2} = x - 9$. Explain what is unique about the solution.

Practice #25 · Work-rate (Kenji + Oscar) Setup: $\frac{1}{a} + \frac{1}{b} = \frac{1}{t}$. How long together?

Practice #33 · PERFORMANCE TASK (★) Chemistry mixture: set $f(x) = \frac{50(0.02) + 0.06x}{50 + x}$ equal to target concentration. Solve for x . Verify with TABLE.

DOK 2

5 min

DOK-3 reveal + LEHS bridge

Sentence frame · Chemistry (#33) “I set up $f(x) = \frac{50 \cdot 0.02 + 0.06x}{50 + x} = 0.05$. After clearing the denominator I got _____. Solving for x gave $x = \frac{\quad}{\quad}$ gallons. I verified with a calculator TABLE by entering $y_1 = \frac{\quad}{\quad}$ and $y_2 = \frac{\quad}{\quad}$.”

Bridge to Topic 4 LEHS Assessment Q#2: domain-restriction reasoning · Q#6: solve rational equation (same moves as #33) · Q#7: work-rate reasoning. You have now rehearsed all three.

DOK 1–2

3 min

not graded

Complete on your exit ticket

1. After clearing denominators I must check each candidate in the _____ equation to catch _____ solutions.
2. In Practice #33 (chemistry), the equation came from setting _____ equal to the target concentration 0.05.
3. One biggest thing I learned today: _____.